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F															
UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-29 & MVD-30 20kV Centralized Panel Room WICA SCHEMATIC DIAGRAM SDS RISER WITH VT H01															
				Date Prep.		15.11.2018		PROJECT TITLE				ABB		EPMV	
				Prepared		YR		MV Switchgear for MNA Serang Plant							
00		Issued for Approval		11.01.19		YR		Checked		AW					
Ind.		Revision		Date		Name		Approved		AI		Based on: BE318043			
												PT. ABB Sakti Industri			
												COVER SHEET		Scale	
												MVD-29 & MVD-30 20kV Centralized Panel Room WICA		Customer	
												SCHEMATIC DIAGRAM		PT WILMAR NABATI INDONESIA & 312	
												SDS RISER WITH VT		Drawing Nbr.	
														IDUNISEC-BE318042-310.10	
														Sh. 1	
														Customer Doc. Nbr.	
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PAGE		PAGE NAME				REVISION									
==MVD-34=H01&312/1		COVER SHEET				00									
==MVD-34=H01&312/2		Index of Sheet				00									
==MVD-34=H01&312/3		REFERENCE DESIGNATIONS				00									
==MVD-34=H01&312/4		REFERENCE DESIGNATIONS				00									
==MVD-34=H01&312/5		REFERENCE DESIGNATIONS				00									
==MVD-34=H01&312/6		REFERENCE DESIGNATIONS				00									
==MVD-34=H01&312/7		DISTRIBUTION OF AUXILIARY CIRCUITS				00									
==MVD-34=H01&312/8		AUXILIARY MOTOR & RELAY				00									
==MVD-34=H01&312/9		SPARE CONTACTS				00									
==MVD-34=H01&312/10		MAIN CIRCUITS				00									
==MVD-34=H01&312/11		LV DOOR LAYOUT				00									
==MVD-34=H01&312/12		LV BOX LAYOUT				00									
==MVD-34=H01&312/13		PART LIST				00									
==MVD-34=H01&312/14		Terminal diagram ==MVD-34=H01+-XDC1													
==MVD-34=H01&312/15		Terminal diagram ==MVD-34=H01+-XDC2													
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==MVD-34=H01&312/18		Terminal diagram ==MVD-34=H01+-XDI6													
==MVD-34=H01&312/19		Terminal diagram ==MVD-34=H01+-XDV													
==MVD-34=H01&312/20		Terminal diagram ==MVD-34=H01+-XDX													

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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE						-QBH	MANUAL CIRCUIT-BREAKERS					
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						-QBL	LINKS					
	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS						-QBM	MINIATURE SWITCH-DISCONNECTORS					
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING						-QBS	SWITCH-DISCONNECTORS					
	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT						-QBT	TRUCK					
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER						-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING					
	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS						-QCE	EARTHING SWITCH					
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER						-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)					
	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER						-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)					
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						-QQ	CLUTCHES					
B	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS						-RAA	ANTIFERRORESONANCE RESISTOR					
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER						-RAD	DIODES					
	-PFB	BLUE SIGNAL LAMPS						-RAI	INDUCTORS					
	-PFF	FLAG RELAYS						-RAR	RESISTORS					
	-PFG	GREEN SIGNAL LAMPS						-RF	FILTERS					
C	-PFR	RED SIGNAL LAMPS						-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING					
	-PFS	SHORT CIRCUIT INDICATORS						-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT					
	-PFV	VOLTAGE INDICATORS						-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE					
	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE						-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION					
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE						-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD					
	-PFW	WHITE SIGNAL LAMPS						-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1					
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS						-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2					
	-PFY	YELLOW SIGNAL LAMPS						-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2					
	-PGA	AMMETERS						-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS					
	-PGC	COUNTERS						-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4					
D	-PGF	FREQUENCYMETERS						-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5					
	-PGH	HOURMETERS						-SFA	AMMETRIC SWITCHES					
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE						-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS					
	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)						-SFL	LOCKING CONTACTS					
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)						-SFO	OPENING PUSH-BUTTONS					
	-PGJ	ACTIVE ENERGY METERS						-SFR	RESET PUSH-BUTTONS					
	-PGK	REACTIVE ENERGY METERS						-SFS	SELECTOR SWITCHES					
	-PGM	MULTIFUNCTION INDICATORS						-SFT	TEST PUSH-BUTTONS					
	-PGP	POWER-FACTOR METERS						-SFU	UNLOCKING PUSH-BUTTONS					
	-PGQ	VARMETERS						-SFV	VOLTMETRIC SWITCHES					
E	-PGS	SYNCHRONOSCOPES						-TA	POWER TRANSFORMERS					
	-PGV	VOLTMETERS						-TB	CONVERTER					
	-PGW	WATTMETERS						-TB1	RECTIFIER FOR FIRST OPENING RELEASE					
	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)						-TB2	RECTIFIER FOR SECOND OPENING RELEASE					
	-QAB	CIRCUIT-BREAKERS						-TB3	RECTIFIER FOR CLOSING RELEASE					
	-QAC	CONTACTORS (FOR POWER)						-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING					
	-QBD	DISCONNECTORS						-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT					
	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE					
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER					
	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)												
F	-QBD4	TWO-WAY DISCONNECTOR												
	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER												

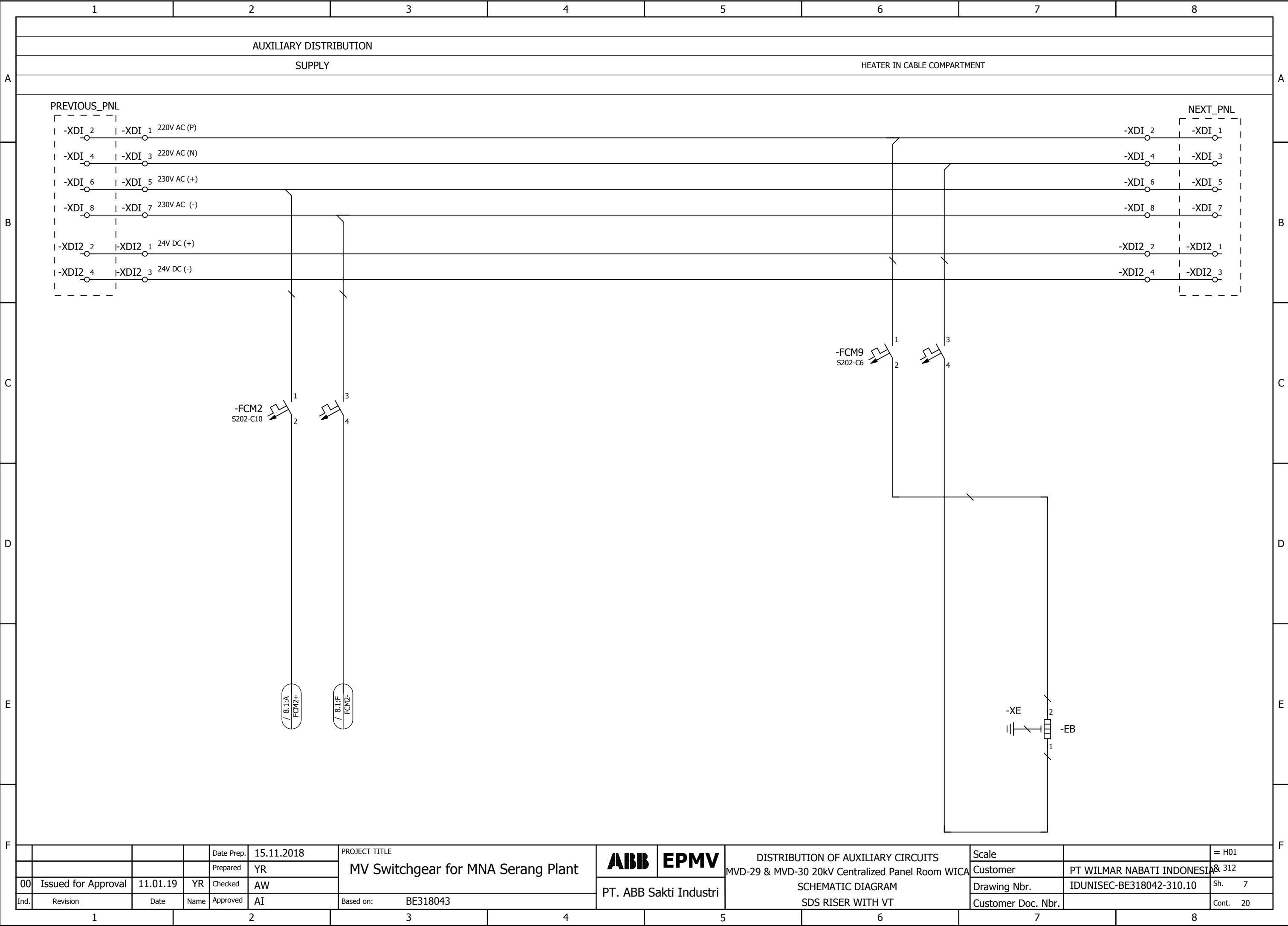
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				Prepared	YR	MV Switchgear for MNA Serang Plant					MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Customer	PT WILMAR NABATI INDONESIA	& 312
00	Issued for Approval	11.01.19	YR	Checked	AW						PT. ABB Sakti Industri	SCHMATIC DIAGRAM	Drawing Nbr.	IDUNISEC-BE318042-310.10
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318043				SDS RISER WITH VT	Customer Doc. Nbr.		Cont. 20	

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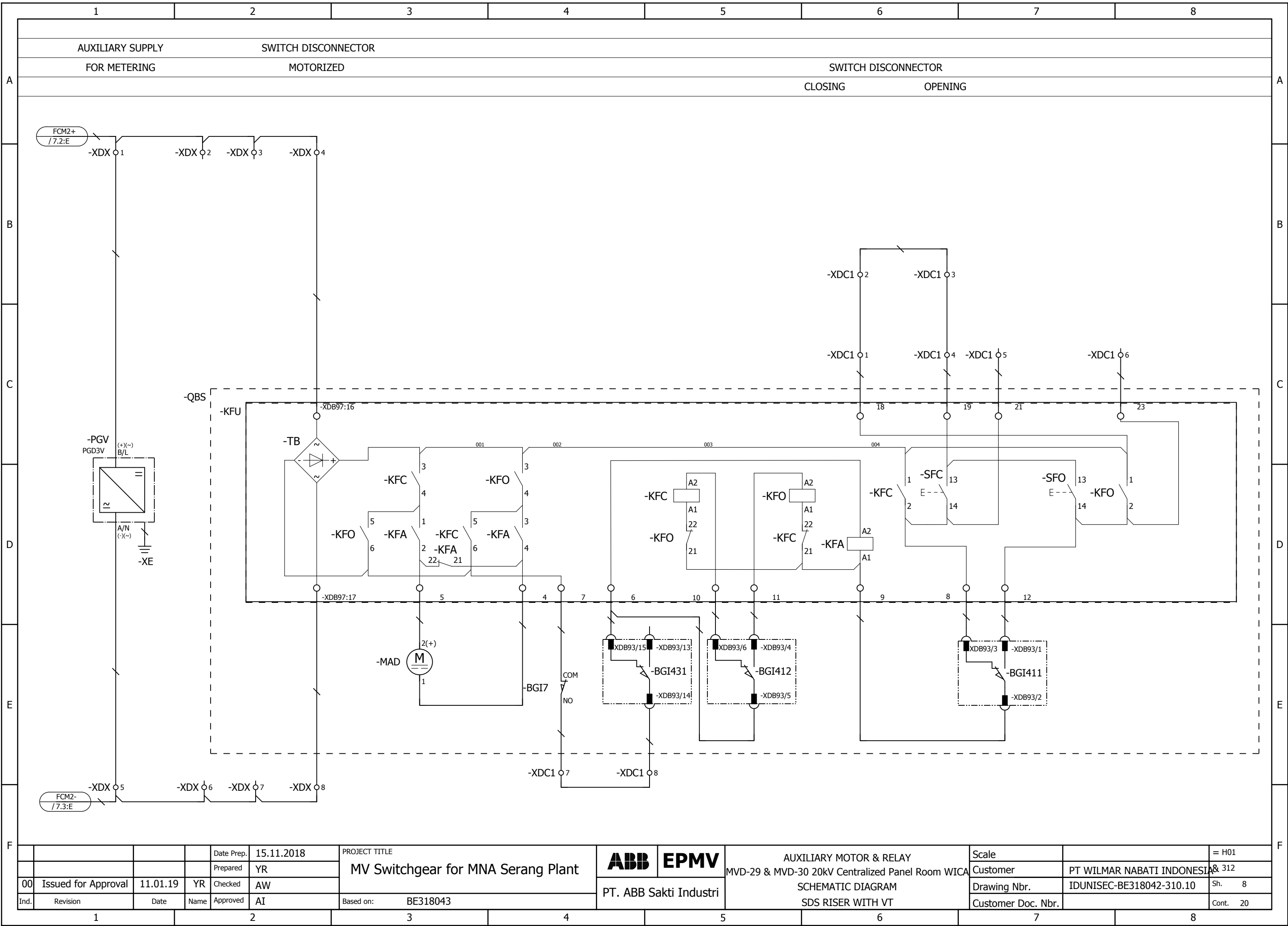
A	-TFA	ACTIVE POWER TRANSDUCERS	A	
	-TFC	CURRENT TRANSDUCERS		
	-TFF	FREQUENCY TRANSDUCERS		
	-TFJ	ACTIVE ENERGY TRANSDUCERS		
	-TFK	REACTIVE ENERGY TRANSDUCERS		
	-TFM	MULTIFUNCTION TRANSDUCERS		
	-TFP	POWER-FACTOR TRANSDUCERS		
	-TFQ	REACTIVE POWER TRANSDUCERS		
	-TFS	SIGNAL CONVERTERS		
	-TFV	VOLTAGE TRANSDUCERS		
B	-WA	BUSBARS	B	
	-WBC	POWER CABLES		
	-WE	EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.		
	-WGC	CONTROL CABLES		
	-WGS	COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)		
	-WH	OPTICAL FIBRES		
	-XDA	TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS		
	-XDA1	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1		
	-XDA2	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2		
	-XDA3	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3		
C	-XDA5	CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER	C	
	-XDB	CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS		
	-XDB1	CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)		
	-XDB2	CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER		
	-XDB10...-XDB89	CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS		
	-XDB7	CONNECTOR FOR TRUCK POSITION CONTACTS		
	-XDB91	CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR		
	-XDB92	CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR		
	-XDB93	CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR		
	-XDB95	CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET		
D	-XDC	CUSTOMER TERMINAL BLOCK	D	
	-XDC1	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER		
	-XDC2	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR		
	-XDC3	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT		
	-XDC4	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS		
	-XDC5	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER		
	-XDE	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE		
	-XDE1	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE		
	-XDE2	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2		
	-XDE3	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2		
E	-XDF	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS	E	
	-XDF1	CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR		
	-XDF2	CONNECTOR FOR REAR VENTILATOR CIRCUITS		
	-XDF3	CONNECTOR FOR FRONT VENTILATOR CIRCUITS		
	-XDH	TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)		
	-XDI	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)		
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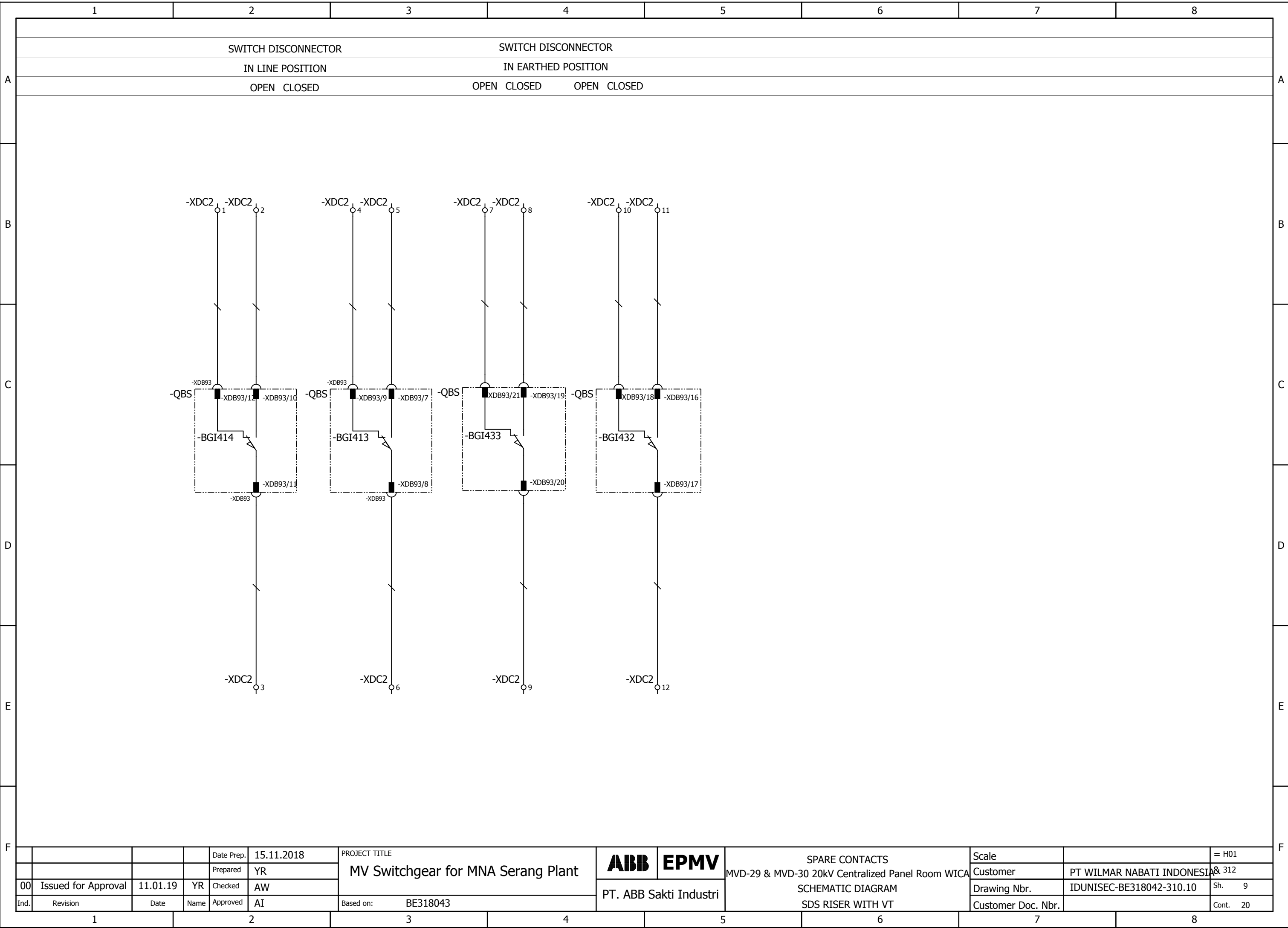


				Date Prep.	15.11.2018	PROJECT TITLE		ABB EPMV	DISTRIBUTION OF AUXILIARY CIRCUITS MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Scale		= H01	
				Prepared	YR	MV Switchgear for MNA Serang Plant				Customer	PT WILMAR NABATI INDONESIA & 312		
00	Issued for Approval	11.01.19	YR	Checked	AW					Drawing Nbr.	IDUNISEC-BE318042-310.10	Sh.	7
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	PT. ABB Sakti Industri	SDS RISER WITH VT	Customer Doc. Nbr.		Cont.	20

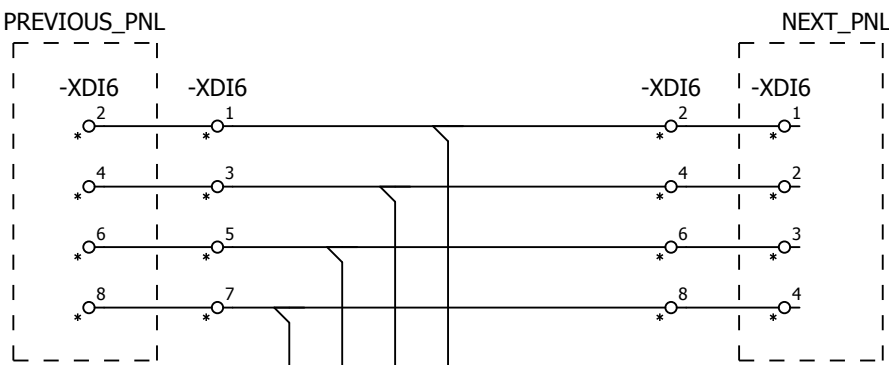
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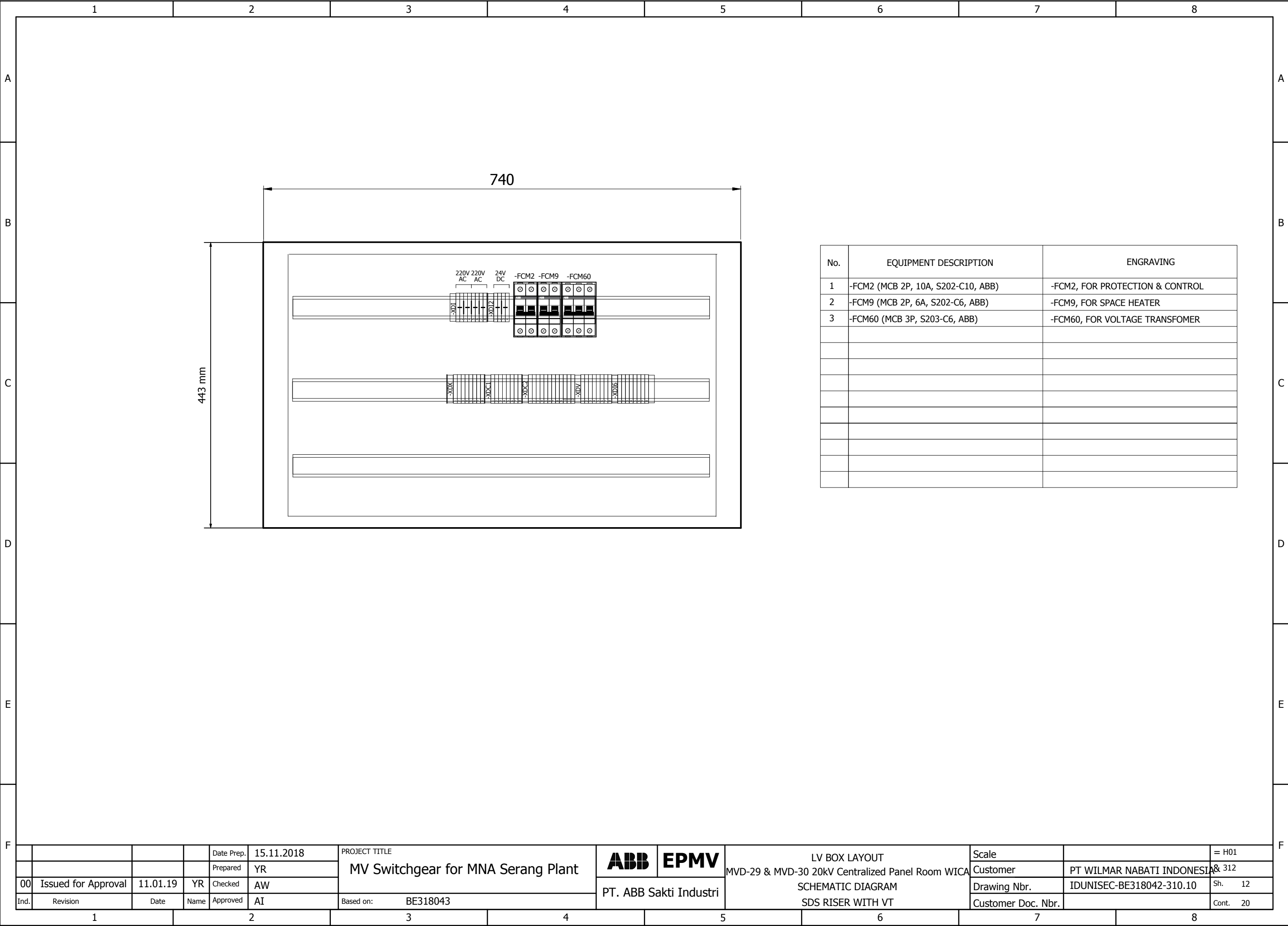
00	Issued for Approval		11.01.19	YR	Checked	AW	PROJECT TITLE	MV Switchgear for MNA Serang Plant	Based on:	BE318043	ABB	EPMV	PT. ABB Sakti Industri	SPARE CONTACTS		MVD-29 & MVD-30 20kV Centralized Panel Room WICA		SCHEMATIC DIAGRAM		SDS RISER WITH VT		Scale				= H01	
	Revision		Date	Name	Approved	AI																Customer		PT WILMAR NABATI INDONESIA		& 312	
																						Drawing Nbr.		IDUNISEC-BE318042-310.10		Sh. 9	
																						Customer Doc. Nbr.				Cont. 20	



				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	MAIN CIRCUITS MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Scale		= H01
				Prepared	YR					Customer	PT WILMAR NABATI INDONESIA	& 312
00	Issued for Approval	11.01.19	YR	Checked	AW			PT. ABB Sakti Industri		SCHEMATIC DIAGRAM	Drawing Nbr.	IDUNISEC-BE318042-310.10
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	PART LIST MVD-29 & MVD-30 20kV Centralized Panel Room WICA SCHEMATIC DIAGRAM SDS RISER WITH VT	Scale		= H01
				Prepared	YR					Customer	PT WILMAR NABATI INDONESIA	& 312
00	Issued for Approval	11.01.19	YR	Checked	AW					Drawing Nbr.	IDUNISEC-BE318042-310.10	Sh. 13
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318043	PT. ABB Sakti Industri		Customer Doc. Nbr.		Cont. 20	

F		E		D		C		B		A								
==MVD-34=H01+-XDC1		REMARK	CABLE	EQUIPMENT	Wire No	Terminal number	TYPE	WIRE NO.	EQUIPMENT	CABLE	REMARK							
		BLACK 1,5 mm²		-QBS-KFU-XDB97:18		1	PT 2.5											
						2	PT 2.5											
						3	PT 2.5											
		BLACK 1,5 mm²		-QBS-KFU-XDB97:19		4	PT 2.5											
		BLACK 1,5 mm²		-QBS-KFU-XDB97:21		5	PT 2.5											
		BLACK 1,5 mm²		-QBS-KFU-XDB97:23		6	PT 2.5											
						7	PT 2.5	-QBS-BGI7:NO				BLACK 1,5 mm²						
						8	PT 2.5	-QBS:-XDB93/14				BLACK 1,5 mm²						
		REMARK: ☆ INTERPANEL LIGHT BUS,OFFERED TOGETHER WITH PANELS. △ OUTGOING CONTROL CABLE (Supply by Customer). ● BRIDGE BAR. ✕ PARTITION PLATE. (TEST JUMPER.																
Ind.		Revision		Date	Name	Approved	Based on:		PROJECT TITLE		PT. ABB Sakti Industri		Terminal diagram ==MVD-34=H01+-XDC1		Scale		= H01	
									MV Switchgear for MNA Serang Plant		ABB EPMV				Customer		PT WILMAR NABATI INDONESIA	
															Drawing Nbr.		Sh. 14	
															Customer Doc. Nbr.		Cont. 20	
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F		E		D		C		B		A														
REMARK		CABLE	EQUIPMENT	Wire No	Terminal number	TYPE	WIRE NO.	EQUIPMENT	CABLE	REMARK														
==MVD-34=H01+-XDI6			-PREVIOUS_PNL-XDI6:2		1 ●	PT 2.5		-PGV:1		BLACK 1,5 mm²														
					2 ●	PT 2.5		-NEXT_PNL-XDI6:1																
				-PREVIOUS_PNL-XDI6:4		3 ●	PT 2.5		-PGV:3		BLACK 1,5 mm²													
					4 ●	PT 2.5		-NEXT_PNL-XDI6:2																
				-PREVIOUS_PNL-XDI6:6		5 ●	PT 2.5		-PGV:5		BLACK 1,5 mm²													
					6 ●	PT 2.5		-NEXT_PNL-XDI6:3																
				-PREVIOUS_PNL-XDI6:8		7 ●	PT 2.5		-PGV:6		BLACK 1,5 mm²													
					8 ●	PT 2.5		-NEXT_PNL-XDI6:4																
		REMARK: ☆ INTERPANEL LIGHT BUS,OFFERED TOGETHER WITH PANELS. △ OUTGOING CONTROL CABLE (Supply by Customer). ● BRIDGE BAR. ✕ PARTITION PLATE. (TEST JUMPER.																						
Ind.		Revision		Date	Name	Approved	Based on:		BE318043		PROJECT TITLE		MV Switchgear for MNA Serang Plant		ABB EPMV		Terminal diagram ==MVD-34=H01+-XDI6		Scale		= H01			
																			Customer		PT WILMAR NABATI INDONESIA		Sh. 18	
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																			Customer Doc. Nbr.					
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